Lukas J. Hondrich

Berlin, Germany

website
lukashondrich@gmail.com

Profile

AI engineer with 7 years in production ML/AI, working across LLM application development, self-hosted inference infrastructure, multi-agent systems, and evaluation, with health and education as core domain areas. Background bridging applied engineering and academic research on human oversight of automated systems.

Selected Projects

MedGraph [GitHub] [Demo] 2026

- a local-inference multi-agent healthcare system for regulated-domain workflows LangGraph parallel fan-out, local OpenAI-compatible inference with Qwen3-8B router + Qwen3.6-27B specialists served via llama-server/llama-swap, FastMCP OpenFDA tool layer, embedded Qdrant RAG, Docker Compose, and Helm/OpenShift-compatible manifests. Validated model routing, safety guardrails, and local inference behavior with benchmark and LLM-judged eval artifacts *LangGraph, Qdrant, Qwen3.6 (self-hosted), FHIR R4*

LLM Oversight Evaluation [Poster] [GitHub] 2026

- Tested whether LLMs systematically omit safety-critical information for non-expert audiences. 40 medication Q&A items × 3 audience conditions × 4 models; 393 FDA-grounded checklist items. 3-judge PoLL validation (κ = 0.784, OpenAI subset). Findings: cross-family audience effect (11.8–16.0pp completeness drops, all p<.001); prompt-based recovery was capability-dependent. Operationalizes the verifiability framework from prior peer-reviewed work. *Python, Inspect AI, Ollama*

Language Tutor [GitHub] [Demo] 2025

- Building open-ended **real-time voice-to-voice** conversational language tutor with persistent learner memory, RAG, and interactive WebGL vocabulary visualization *Python, OpenAI, Haystack*

Professional Experience

Independent AI Engineer 09, 2025 – present
Berlin, Germany

- **Apple – Senior Data Analyst**
  AI Safety
- ***Open Working Hours***: an Android & iOS app, live on the App Store, for anonymous reporting and analysis of overtime in healthcare. Differential privacy, k-anonymity. Privacy infrastructure extracted as PyPI-package [dp-group-stats] *React Native, FastAPI, PostgreSQL, Docker*
- Building **agentic document-extraction system** for a German industrial equipment manufacturer: parsing customer inquiries (PDFs, emails) into structured parameters with confidence scores and human-review loop, EU-resident deployment *Python, Qwen3.5, Mistral OCR, Scaleway*

AI Engineer - Quality & Safety 02, 2024 – 08, 2025
Cornelsen - educational technology *Berlin, Germany*

- Designed and implemented a **Quality and Safety platform** for running automated evaluations, data-driven analyses, resulting in significant improvements in error rates *Python, Streamlit*
- Conducted **statistical modelling** using linear mixed-effects models to analyze temporal trends in conversation flows helping domain experts improve AI quality *Python, SciPy*

- **Student space modelling**, using psychological, demographic, didactic dimensions to generate coverage optimized synthetic data on demand *Python*
- **Led development** of a multilingual (Llama3.3) tutoring app in a highly regulated educational setting for 25k students, on market within 10 weeks *Python, FastAPI*
- **Optimized system performance** through function calling and rate limiting, reducing inference costs by 50% and increasing concurrent capacity to 520 users (2.4x) *Python, FastAPI*
- **Optimized AI/RAG** for three customer-facing educational applications leading to significant improvements in retrieval quality and customer satisfaction *Python, FastAPI, ChromaDB*
- Engineered **PII-removal service**, including finetuning of Llama3.1, PEFT/LoRA, integrated with Presidio *FastAPI, LoRA, Presidio, Docker, K8s*
- Cornelsen.ai was awarded the **Comenius EduMedia Medal 2025** (highest distinction for digital educational media)

Researcher (Part-time) 10, 2022 – 03, 2024
University Hagen *Hagen, Germany*

- Reviewed and developed technical architectures for human oversight in LLM-based systems, ensuring quality, safety, and compliance
- Co-authored two peer-reviewed publications on AI governance and oversight for automated decision-making systems

AI Engineer 03, 2022 – 01, 2024
Institute of Medical and Human Genetics, Charité / LIMAA Technologies *Berlin, Germany*

- Built the **computer-vision pipeline** for a light-sheet microscopy platform — 3D segmentation (cVAE, U-Net) detecting tissue structures in cancer samples (colon / small intestine), with expert-in-the-loop refinement *Python, TensorFlow, OpenCV*
- Built Infrastructure for **deploying, logging, analyzing** experiments *Bash, Slurm, Neptune.AI*
- **Created Researcher productivity tool**: iOS app for processing of 3D images *Python, PyQT, PySide*

Research Assistant 03, 2021 – 03, 2022
AlgorithmWatch *Berlin, Germany*

- **Consulted** on measuring and improving fairness in embedding-based search systems
- **Advised Bosch GmbH** on implementing AI fairness measures in production systems

Machine Learning Engineer 10, 2018 – 09, 2020
Thomann - e-commerce *Berlin, Germany*

- Built collective filtering based **recommendation engine** *GCP, Tensorflow, PySpark, Python, SciPy*
- Built **data analysis tool** for search engine usage for >1 million daily customers, significantly decreasing drop-off rate *GCP, Docker, Python, SciPy*
- Engineered forecasting system for global parcel delivery *GCP, Docker, Python, TF*

Selected Technical Skills

LLM & Alignment:

- Context Engineering, Safety Evals, PEFT/LoRA finetuning, RLHF, Ollama, LangGraph, LLamaIndex, LiteLLM, LangFuse, Gemma4, Llama3.3, Qwen3, llama.cpp, FastMCP, llama-swap

Machine Learning & Data

- *Python, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, mongoDB, chromaDB, Elasticsearch, Haystack*

APIs & Infrastructure

- *OpenAI API, Huggingface Transformers, Accelerate, FastAPI, Docker, Kubernetes, Terraform, GCP, Azure, Hetzner, Helm, Cloudflare Tunnel*

Tools & Collaboration

- *GitHub, CI/CD pipelines, Playwright, Claude Code, OpenAI Codex, Hermes*

Education

DPhil in Computer Science (offer declined) 2023
Oxford University *Proposal: Participatory Governance for RL Systems*

M.Sc. Cognitive-Affective Neuroscience 2015 – 2018
Technical University Dresden *grade: 1.6*

B.Sc. Psychology 2011 – 2015
Johannes Gutenberg-Universität Mainz *grade: 1.7*

Other Research Experience

Internship, M.Sc. Thesis 04, 2017 – 04, 2018
Bernstein Center for Computational Neuroscience & Princeton University *Berlin, Germany*

- **Thesis (grade 1.1)**: attention & learning in multidimensional environments *1.1*
- Modelled behavioural data with **reinforcement-learning & Bayesian learning** *Matlab, Slurm*
- Performed **Bayesian model comparison** on high-dimensional fMRI data

Workshops, Summer-Schools, Courses

AI-Officer Certification 12, 2024
aiofficer.de, according to Art. 4, EU-AI-Act *Berlin, Germany*

- Training: Building AI-systems compliant to the EU-AI-Act

The Race to Regulate AI: Global Comparative Perspectives 06, 2022
Oxford University *Oxford, UK*

- Workshop: EU-AI-act: High-risk systems and protection of fundamental rights

Relevant Publications

Implementing employee interest along the machine learning pipeline
Hondrich, Mollen, Ponce del Castillo (ed.), ETUI, *Artificial intelligence, labour and society (2024)* [paper]

Automation Bias in public administration – an interdisciplinary perspective from law and psychology
Ruschemeier, **Hondrich** (2024). Government Information Quarterly. [paper]

Three-Dimensional Histological Characterization of the Placental Vasculature Using Light Sheet Microscopy
Freise, **Hondrich**, Hägerling et al. (2023). *Biomolecules* [paper]

Addressing Automation Bias through Verifiability
Hondrich, Ruschemeier (2023). *European Conference on Alg. Fairness.* [paper]

Working paper: From risk mitigation to employee action along the machine learning pipeline
Mollen, **Hondrich** (2023). *Böckler Impuls*. [paper]

Public Speaking

Evaluating Multi-turn Conversational AI — *EvalOps unfiltered: Evaluating LLM-based applications*
Merantix AI Campus — *2025*

Shaping a new social contract for AI — *Panel: Book launch of "Artificial intelligence, labour and society"*
European Trade Union Institute — *2024*

EWAF 2023 - Lightning Round — *Verifiability as a Minimal Requirement for Human Oversight*
Europ. Workshop on Alg. Fairness — *2023*

AI Talks at ETUI — *The Machine Learning Pipeline: Space for Action*
European Trade Union Institute — *2022*

Languages

German - native
English - C1
Spanish - B1